→ From notes, we know the joint posterior

$$p(\mu, \sigma^2 \mid \underset{\sim}{y}) \propto (\sigma^2)^{-\frac{n}{2}-1} \exp\left\{ \frac{-1}{2\sigma^2} \left[(n-1)s^2 + n(\bar{y}-\mu)^2 \right] \right\}$$

→ Using Gibbs sampler, we look at the distribution of each parameter, conditioned on all the others

$$\Rightarrow \quad p(\mu \mid \sigma^2, y) \propto \exp\left\{ \frac{-n}{2\sigma^2} (\bar{y}-\mu)^2 \right\}$$

$$= \exp\left\{ \frac{-(\mu-\bar{y})^2}{2(\sigma^2/n)} \right\}$$

$$\downarrow$$

$$\sim \text{Normal}\left(\bar{y}, \frac{\sigma^2}{n}\right)$$

$$\& \quad p(\sigma^2 \mid \mu, \underset{\sim}{y}) \propto (\sigma^2)^{-(\frac{n}{2}+1)} \exp\left\{ \frac{-1}{2\sigma^2} \left[(n-1)s^2 + n(\bar{y}-\mu)^2 \right] \right\}$$

< nothing drops out & rewrote first exponent to match form of inv-Gamma (α, β) >

$$\downarrow$$

$$\sim \text{Inv-Gamma}\left(\alpha = \frac{n}{2}, \beta = (n-1)s^2 + n(\bar{y}-\mu)^2\right)$$